

Course Code: CS1103		
Name of the Programme: BCA (Hons.)		
Name of the Course: Operating Systems		
Semester: II		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
3	2:0:2	60
Course Objectives:		
a) To equip students with a strong foundational knowledge of OS concepts.		
b) To covers key components of an operating system including Process management Memory management, Deadlock detection and handling, Device and I/O management.		
c) To introduce students about Advanced optimization techniques, Scheduling algorithms and mechanisms, System-level resource management.		
d) To emphasize both theoretical understanding and practical application.		

Course Outline (Syllabus of the Course)

Module - 1 (Basics of operating systems, Process management)	No. of Hours
	6 Hrs.
Generations, Types, Structure, Services, System Boot, System Programs, Protection and Security. Process Concepts, Process States, Process Control Block, Scheduling-Criteria, Scheduling Algorithms and their Evaluation.	
Module - 2 (Thread API)	No. of Hours
	6 Hrs.
Thread Creation, Completion, Locks, Condition Variables, Compilation. Inter Process Communication and Synchronization: Software Approaches, Principles of Concurrency, Hardware Support, Semaphores, Message Passing.	
Module - 3 (System Calls, Traps, Device Drivers, and Deadlocks)	No. of Hours
	6 Hrs.
System Calls – Concepts and Execution Flow, User Mode and Kernel Mode, Trap Handling and Exception Flow, Interrupt Service Routines (ISRs), Implementation of System Calls in xv6, Device Communication and Drivers - Deadlocks – Characterization, Deadlocks in I/O and Device Handling, Prevention, Detection, and Avoidance Techniques in Kernel-Level Resource Management.	

Module - 4 (Memory Management)	No. of Hours
	6 Hrs.
Main Memory, Swapping, Contiguous Memory Allocation, Paging, Structure of Page Table, Segmentation, Virtual Memory, Demand Paging, Page Replacement Algorithms, Allocation of Frames, Thrashing.	
Module - 5 (Storage Structure)	No. of Hours
	6 Hrs.
Overview of Mass Storage Structure, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, I/O System Overview, I/O Hardware, Application I/O Interface.	

Course outcomes :
a) Understanding of the historical evolution and basic principles of operating systems.
b) Understand the process management techniques and scheduling.
c) Analyze deadlock conditions and propose methods to avoid or resolve them.
d) Analyze and address issues such as memory fragmentation and dynamic memory allocation.

Text Books:	
1	Stallings, W. (2018). <i>Operating systems: Internals and design principles</i> (9th ed.). Pearson Education. ISBN 978-9352866717
2	Silberschatz, A., Galvin, P. B., & Gagne, G. (2010). <i>Operating system concepts essentials</i> . John Wiley & Sons. ISBN: 978-0470889206
3	Cox, R., Kaashoek, F., & Morris, R. (2022). <i>xv6: A simple, Unix-like teaching operating system</i> (Rev. 3, RISC-V ed.). MIT CSAIL.

Reference Books:	
1	Silberschatz, A., Galvin, P. B., & Gagne, G. (2012). <i>Operating system concepts</i> (9th ed.). Wiley. ISBN: 978-1118063330
2	Tanenbaum, A. S., & Bos, H. (2023). <i>Modern operating systems</i> (Global 5th ed.). Pearson. ISBN: 978-1292459660

Laboratory Components	
1	Process Creation in Linux: Creating Child Processes and Displaying Parent-Child Relationships Using ps.
2	Process Control: Create a shell script that demonstrates process control commands like bg, fg, jobs, and kill. Implement a simple background task and manage its execution.
3	Basic Shell Script for Job Scheduling: Develop a shell script that takes a list of commands and executes them with a specified delay using the sleep command. Demonstrate job scheduling in the shell.
4	File Operations: Create a shell script that performs file operations, such as creating, copying, moving, and deleting files using commands like touch, cp, mv, and rm. Include error handling to manage file operations gracefully.
5	File Permissions: Write a shell script to modify file permissions and ownership using chmod and chown. Include user input for the filename and desired permissions and display the new file permissions.
6	Directory Management: Develop a shell script that creates, lists, and deletes directories. Use commands like mkdir, ls, and rmdir, and include user prompts for directory names.
7	Dynamic Memory Check: Write a shell script that monitors memory usage on the system using commands like free and top. Display the current memory usage and available memory.
8	Environment Variables: Create a shell script that sets, exports, and displays environment variables. Use printenv and env to show the current environment settings.
9	I/O Redirection in Unix Shell: Manipulating Data Streams with Enhanced File Operations.
10	Command Piping in Linux: Exploring Operating System Features.

LESSON PLAN

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Name of the Course	Operating Systems	
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Course Credit	No. of Hours per Week (L+T+P)	Total Teaching Hours
3	2:0:2	60
Course Instructor	Dr. K. Thenmozhi and Dr. Mahadev Patil	

Session	Module	Topic / Experiment	Pedagogy / Activity	Reference / Tools
1	1	Introduction to Operating Systems	Lecture	Stallings
2		Generations and Types of Operating Systems	Lecture	Stallings
3		Process Creation in Linux	Hands-on	Linux
4		Parent–Child Process Observation	Practice	Linux
5		Structure, Services and security	Lecture	Stallings
6		Process concepts and states, PCB	Lecture	Stallings
7		Process Control (bg, fg, jobs)	Hands-on	Shell
8		kill command usage	Practice	Shell
9		Scheduling criteria and algorithm	Lecture	Silberschatz
10		Algorithms and evaluation	Lecture	Silberschatz
11		Shell Script – Job Scheduling	Hands-on	Bash
12		sleep-based scheduling	Practice	Bash
13		Thread Creation, completion	Lecture	Silberschatz

14	2	Locks , condition variables and compilation	Lecture	Silberschatz
15		File Operations Script	Hands-on	Shell
16		Error Handling in Files	Practice	Shell
17		IPC	Lecture	Silberschatz
18		Synchronization	Lecture	Silberschatz
19		File Permissions using chmod	Hands-on	Linux
20		Ownership using chown	Practice	Linux
21		Software approaches and principles of concurrency	Lecture	Silberschatz
22		Semaphores and Message passing	Lecture	Silberschatz
23		Directory Management	Hands-on	Linux
24		Directory listing and deletion	Practice	Linux
25	3	System calls and execution flow	Lecture	Silberschatz
26		Mode of Os ISR	Problem Solving	Silberschatz
27		Environmental Variables	Hands-on	Shell
28		export and printenv	Practice	Shell
29		Implementation of system call	Lecture	Silberschatz
30		Device communication and drivers, Deadlocks	Lecture	Silberschatz
31		I/O Redirection	Hands-on	Shell
32		Input and Output Streams	Practice	Shell
33		Deadlock prevention and detection	Lecture	Silberschatz
34		Deadlock Avoidance techniques	Lecture	Silberschatz
35		Command Piping	Hands-on	Shell
36		Multiple Command Pipelines	Practice	Shell
37		Introduction of memory and Swapping	Lecture	Silberschatz
38		CMA and paging	Lecture	Silberschatz
39		System Call Observation	Demo	strace

40	4	Trap and Interrupt Analysis	Practice	Linux
41		Paging continuation and page Table	Lecture	Silberschatz
42		Segmentation and virtual memory	Lecture	Silberschatz
43		Memory Usage Monitoring	Hands-on	free
44		top command analysis	Practice	Linux
45		Demand paging and page replacement	Lecture	xv6
46		Page replacement algorithm and Thrashing	Lecture	xv6
47		Paging Observation	Practice	Linux
48		Virtual Memory Concepts	Demo	Linux
49		Introduction to storage structure	Lecture	Silberschatz
50	5	Disk structure and scheduling	Lecture	Silberschatz
51		Disk Scheduling Demo	Case Study	Tool
52		Disk Access Analysis-1	Practice	Tool
53		Disk Scheduling and Management	Lecture	Silberschatz
54		Swap space management and I/O system overview	Lecture	Silberschatz
55		Disk Access Analysis-2	Practice	Tool
56		Viva Voce	Demo	-
57		I/O Hardware and Interface	Lecture	Silberschatz
58		Recap of all Modules	Lecture and Discussion	-
59		Lab Test	Assessment	-
60		Lab Test Continuation	Assessment	-

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 1. Quiz 2. Problem & Scenario Based Assessment (pen and paper)	10 marks out of 70 10 marks out of 70
CIE-2 3. Written Test (Pen and Paper)	25 marks out of 70
CIE-3 4. GitHub activity log / GCR 5. Lab Test + Viva voce	5 marks out of 70 20 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

PO/CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO 12
CO1	3	2	1	2	-	-	-	1	-	-	-	-
CO2	3	2	2	3	1	-	-	-	-	-	-	-
CO3	3	3	3	2	-	-	1	-	-	-	-	-
CO4	3	2	2	3	2	1	-	1	-	-	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks	70
CIE components	Continuous Internal Evaluation -1 (20 Marks) Rubrics for Assessment
CIE-1 Component 1- Quiz Weightage: 10 Marks Component 2 - Problem & Scenario Based Assessment Weightage: 10 Marks	Quiz will be conducted for 10 marks. Totally 10 questions each question carries 1 mark. No negative marking for wrong answers. • This assessment will be conducted for 10 marks. • The question paper consists of 2 questions. Each question carries 5 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty • There will be 4 sets of question papers. • No negative marking for wrong answers.
CIE components	Continuous Internal Evaluation-2 (25 Marks) Rubrics for Assessment
CIE-2 Mid Sem Examination (Pen and Paper) Weightage: 25	The test will be conducted with 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.
CIE components	Continuous Internal Evaluation-3 (25 Marks) Rubrics for Assessment
CIE-3 Component 4 - GitHub Activity log / GCR Weightage: 5	Refer Table 1
CIE-3 Component 5 – Lab Test + Viva Weightage: 20	Refer Table 2

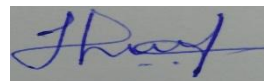
Table 1: Component 4 - GitHub Activity log / GCR

GitHub Activity log / GCR (5 Marks)				
Excellent (5)	Very Good (4)	Good (3)	Average (1-2)	Needs Improvement (0)
Command is fully correct, optimized and well-commented. Handles all the test cases and on time completion	Minor mistakes or inefficiencies. The logic is mostly correct. Minor test cases missed.	Basic logic implemented. Some errors or incomplete handling.	Major logic flaws. Code / command partially runs but is incomplete or error prone.	No submission or irrelevant work

Table 2: Component 5 – Lab Test + Viva

Write-up and Execution rubrics (Max: 15 marks)				
Sl. No	Criteria	Excellent	Good	Poor
1	Understanding the concept and writing the Program/commands for the problem (5 Marks)	Understanding of requirements and suitable Program/commands written for the problem (5 M)	Understanding of requirements and partially written the suitable Program / commands for the problem. (4-1 M)	Do not have a clear understanding of requirements and are unable to write any Program/commands for the problem. (0 M)
2	Execution (5 marks)	Demonstrate the execution of the commands with error correction and show performance efficiency. Appropriate validations with all test cases are handled. (5 M)	Demonstrate the execution of the commands program without error correction of the code and show performance efficiency with only few test cases. (4-1 M)	Students have not executed the program/command. (0 M)

3	Results (5 Marks)	Optimal output and appropriate validations with all test cases are handled (5 M)	Partial output and appropriate validations with few test cases are handled (4-1 M)	No output and not handling any test cases (0 M)
Viva Voce Rubrics (Max: 5 marks)				
1	Conceptual Understanding (3 Marks)	Explain thoroughly the commands executed along with related concepts. (3 M)	Adequately explains the commands executed along with related concepts (2-1 M)	Unable to explain the commands executed along with related concepts (0 M)
2	Communication of concepts (2 Marks)	Communicates the concept of commands used in problem solving well. (2 M)	Sufficiently communicates the concepts used in problem solving. (1 M)	Unable to communicate the concepts used in the problem. (0 M)



Signature of the Faculty

Dr. K. Thenmozhi